

Hassan Khalid

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SUMMARY

AI engineer with hands-on experience in **backend development, LLM evaluation, and data analysis**, passionate about building scalable, real-world applications.

EDUCATION

Forman Christian College University

Bachelor of Science in Computer Science

June 2026

EXPERIENCE

Single Solution

Jr. AI Engineer

Jan 2026 – Mar 2026

- Analyzed and resolved real-world software engineering issues from **open-source GitHub repositories**, validating fixes against automated test suites.
- Contributed to **LLM performance benchmarking** workflows by evaluating **AI-generated code** solutions across large codebases.
- Collaborated with cross-functional teams to identify patterns in AI model failures, providing actionable insights to improve code generation accuracy and reliability.

ACADEMIC PROJECTS

FYP - AI Learning Platform

- Developed an AI-powered learning system with dynamic quiz generation using **FastAPI**, **Next.js**, and **PostgreSQL**.
- Implemented **adaptive assessment** workflows, **JWT authentication**, and **async SQLAlchemy** for a responsive student experience.
- Delivered a modular platform for personalized coding education and scalable API-driven deployment.

Social Media Backend API

- Engineered a social media backend using **FastAPI** with full CRUD functionality and JWT-based authentication for secure user access control.
- Architected a **PostgreSQL** database using **SQLAlchemy ORM** and **Alembic**, implementing a post-voting system with composite primary keys.
- Built a **CI/CD pipeline** using **GitHub Actions** and **Docker**, automating integration tests via **Pytest** and production deployment.

Algorithms Implemented from Scratch

- AI Algorithms: **Linear Regression**, **Logistic Regression**, **Decision Trees**, **K-means Clustering**, SVM Sort
Algorithm: **Bubble Sort**, **Insertion Sort**, **Merge Sort**, Quick Sort Search Algorithms: **Breadth First**, **Depth First**, A-Star Other: **Huffman Encoding**, **Minimax**.

SKILLS

- Programming Languages:** Python, SQL, C++
- Frameworks/Libraries:** NumPy, Pandas, Scikit-learn, Matplotlib, FastAPI, SQLAlchemy, Alembic
- Tools:** Git/GitHub, GitHub Actions, Jupyter Notebook, VS Code, Cursor
- Databases:** SQL Server, MySQL, PostgreSQL